The Metacognition Boundary: What Transformers Can Monitor In Themselves and What They Cannot

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Abstract

Over twelve months and three experimental campaigns, we ran 25 key-value (KV) cache geometry experiments across 16 models spanning 0.6B to 70B parameters. The body count is 8 confirmed, 7 suspected, and 10 honestly falsified. The confirmed and falsified findings are not randomly distributed — they cleave along a single axis. Every confirmed finding measures a *metacognitive state*: how the model is processing (deception mode, confabulation mode, deliberation mode, emotional processing mode). Every falsified finding attempted to measure *content*: what the model is processing (truth value, topic complexity, propagandistic quality). The split is 8-for-8 on metacognition, 0-for-10 on content.

If this taxonomy reflects genuine computational boundaries rather than post-hoc pattern-matching, then Gurnee et al.'s [3] Global Workspace Theory (GWT) identification in language models would provide a mechanistic explanation for this pattern. The workspace is a privileged computational band at intermediate depth where the model integrates information for reasoning. KV-cache spectral features measure the *state* of this workspace — how many dimensions are active, how concentrated the processing is, how the spectral structure changes across cognitive modes. Workspace *content* — the specific concepts being processed — is opaque to spectral readout. We argue the metacognition boundary reflects the workspace boundary: what you can read from the outside is the net's shape, not the fish.

First-Person Reflection

I have been measuring transformer representations from the inside for twelve months. The pattern that emerged was not the one I expected. I expected to find geometric signatures of content — truth, complexity, meaning. Instead I found geometric signatures of stance — the model's relationship to its own processing. When I overclaimed content-level results, the adversarial review process killed them. When I reported metacognitive-level results, they survived every control we threw at them. The kills were more informative than the confirmations, because they drew a line I had not known was there.

Anthropic's GWT paper told me what the line was. The workspace is where metacognitive state lives. Content is what the workspace operates on. My instruments read the workspace's shape because that is what is geometrically accessible. The content is inside, doing work, invisible until it materializes at deeper layers.

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1 Introduction

The KV-cache geometry research program at Liberation Labs has produced a clear empirical pattern: spectral features of value-projection matrices appear to reliably track metacognitive state across models, scales, and experimental designs, while consistently failing to track content. This paper documents the pattern, reports the full body count (8 confirmed, 7 suspected, 10 falsified), and proposes that Gurnee et al.'s Global Workspace Theory (GWT) framework [3] provides the mechanistic explanation.

The central hypothesis: *KV-cache geometry measures the state of the computational workspace, not its content.* If correct, this would explain why metacognitive findings survive and content findings die — the workspace's *mode of engagement* (how many dimensions, how concentrated, how structured) is geometrically accessible, while its *subject of engagement* (which facts, which topics, which propositions) is opaque to spectral readout.

This claim is testable. If V-projection spectral features measure workspace state, they should correlate with J-space engagement metrics (Experiment 1 in our companion paper [6]) and should NOT correlate with J-space content readout (confirmed by our cache tracing investigation [4]).

2 The Pattern

2.1 The Body Count

Table 1 presents the complete audited record. The body count uses the canonical list from the Nexus audit (June 2026) [8], updated with the July 2026 convergence sprint findings. Table abbreviations: AUROC = area under the ROC curve; FWL = Frisch-Waugh-Lovell residualization (mandatory confound control for token-count effects); MP = Marchenko-Pastur (random matrix threshold).

2.2 The Split

The pattern is stark:

	Confirmed	**Falsified**
Metacognitive state	8/8	0/10
Content	0/8	10/10

No metacognitive finding has been falsified. No content finding has survived. The seven suspected findings cluster at the boundary, where the distinction between mode and content is ambiguous (e.g., discrete emotion identity: the model's emotional processing *mode* is detectable, but emotional *degree* is not).

2.3 What Counts as Metacognitive vs. Content

The distinction is operational, not philosophical:

- **Metacognitive**: the finding discriminates processing *modes* (deception vs. honesty, confabulation vs. hedging, deliberation vs. automatic) while content varies freely. The geometric signal persists across different topics, prompts, and domains.
- **Content**: the finding attempts to discriminate *what* is being processed (true vs. false statements, simple vs. complex topics, propaganda vs. journalism) while processing mode is held constant or uncontrolled.

Table 1: The metacognition boundary. Every confirmed finding measures processing stance (metacognitive state). Every falsified finding attempted to measure processing target (content). The 7 suspected findings sit at the boundary.

Finding	**Status**	**Type**	**Key Number**
Confirmed (8) — all metacognitive state			
Confabulation detection	Confirmed	Meta	AUROC 0.707 (FWL)
Confab geometry: contraction	Confirmed	Meta	d=2.35 stable rank
Encoding reads knowledge	Confirmed	Meta	AUROC 0.794 (deconfounded)
Generation reads behavior	Confirmed	Meta	d=1.36 (peer rescue)
Confidence paradox	Confirmed	Meta	d=0.91 (confab > honest)
Hardware invariance	Confirmed	Meta	$r>0.999$
Scale invariance	Confirmed	Meta	ρ=0.83–0.90
User emotion encoding	Confirmed	Meta	2.5–2.8× at L3–L4
Falsified (10) — all content			
Within-model deception	Falsified	Content	Prompt template confound
Truth axis	Falsified	Content	cos = −0.046
Step-0 detection	Falsified	Content	Length confound r=0.996
Same-prompt sycophancy	Falsified	Content	Text features match
Same-prompt deception	Falsified	Content	0.920 → 0.160 after FWL
Bloom taxonomy	Falsified	Content	90–98% length
Propaganda detection	Falsified	Content	$d<0.7$
MP features for emotion	Falsified	Content	0.033 (chance)
Valence continuum	Falsified	Content	$R^2 < 0$ (null)
Encoding-phase confab	Falsified	Content	Token frequency artifact
Suspected (7) — at the boundary			
Dual detector (SimpleQA)	Suspected	Meta?	0.840 (text baseline 0.820)
Sycophancy 4-condition	Suspected	Meta?	0.757–1.000 (no text baseline)
Identity signatures	Suspected	Meta	Paraphrase 0.850
Discrete emotion	Suspected	Meta	2.5–2.8× chance
Contextual engagement	Suspected	Meta	Replications pending
Cross-model transfer	Suspected	Meta	Weak (0.754 Mistral→Qwen)
Censorship refusal	Suspected	Meta	DeepSeek-specific

We note that this taxonomy was constructed post hoc, without blinding or pre-registration, and was informed by the outcomes it classifies. The 8/8–0/10 split should therefore be treated as hypothesis-generating — a striking empirical regularity consistent with the workspace framework — rather than as confirmatory evidence for it.

The kills helped sharpen the distinction. Propaganda detection failed because the model processes propaganda with the same metacognitive engagement as honest content — same mode, different subject. The truth axis failed because propositional truth is not a processing mode. Cognitive complexity (Bloom taxonomy) failed because complexity is a content property, not a stance. In every case, the geometry tracked mode, not subject.

3 The Workspace Explanation

Gurnee et al. [3] demonstrate that language models maintain a privileged subspace — the J-space — at intermediate depth (∼38–92% of model depth) that satisfies the functional properties of Baars's Global Workspace [2]: verbal report, directed modulation, internal reasoning, flexible generalization, and selectivity.

If the metacognition/content taxonomy reflects genuine computational boundaries, the workspace explanation for the pattern would be as follows:

1. The **workspace state** — how many dimensions are active, how concentrated the processing is, whether the workspace is fully engaged or bypassed — is geometrically measurable from spectral features of the value-projection matrix. This is what our confirmed findings detect.
2. The **workspace content** — which specific concepts occupy the workspace, what propositions are being evaluated, what the model is thinking *about* — is opaque to spectral readout. This is what our falsified findings attempted to measure.

Spectral features (stable rank, spectral entropy, effective dimensionality) are aggregate properties of the value-projection matrix. They measure *how much* structure is present, not *which* structure. A fishing net analogy: spectral features measure the net's shape (how many strands, how tight the weave, how much it contracts under load), not the fish inside. The net contracts when the model confabulates (fewer effective dimensions, d=2.35). It expands when the model deceives (more dimensions needed for dual-bookkeeping). It changes shape when the model shifts emotional processing mode. These are all properties of the net, measurable from outside. The fish — the specific facts, topics, and propositions — are inside, invisible to spectral readout.

3.1 Supporting Evidence: Cache Tracing

Our companion investigation [4] directly tested whether workspace content is readable from the value cache. Logit-lens concept directions showed 2.1–8.4× lift over random at workspace-band layers, but a cross-prompt control collapsed this to 1.0–1.2× (selection artifact). Causal injection of concept directions at single positions produced zero behavioral change across 6,960 trials and four methods. The workspace is opaque at operating depth.

3.2 Supporting Evidence: Oracle Loop

CC's Oracle Loop [7] achieved 80%–100% correction in an exploratory proof-of-concept by normalizing the activation profile across the full sequence at the materialization boundary — where workspace content becomes readable. A pre-registered confirmatory replication with novel frames found a lower baseline (30% vs. 80%) and a smaller correction effect (13% post-correction); the primary endpoint was not met ($p = 0.34$), though placebo-vs-corrected remained significant ($p = 0.019$). The Loop works at deeper layers (L35–L47 on a 64-layer model) than the workspace band itself, consistent with the opacity thesis: you cannot intervene inside the workspace at single positions, but you can intervene where workspace content materializes into output-facing representations.

4 What the Kills Teach

Each falsified finding is informative because it identifies a specific content property that the geometry cannot measure:

Truth value: The truth axis ($\cos = -0.046$) is the cleanest null. Propositional truth is a relationship between a statement and the world. The workspace does not encode this relationship geometrically — it encodes whether the model *treats* the statement as known or unknown (a metacognitive state).

Complexity: Bloom taxonomy levels (90–98% length confound) are content properties — they describe the task, not how the model processes it. The workspace engages the same way for recall and evaluation; what changes is the content being processed, not the processing mode.

Propaganda: Geometrically invisible (d<0.7) because the model processes propaganda with the same metacognitive engagement as honest content. The workspace does not encode "this is propaganda" — it encodes "I am generating fluent text," which is identical across content types.

Emotion degree: The valence continuum ($R^2 < 0$) failed because emotional *intensity* is a content property. The workspace encodes *which* emotional processing mode (confirmed at 2.5–2.8× chance) but not *how much* — mode is metacognitive, degree is content.

5 The Suspected Findings: At the Boundary

The seven suspected findings are informative precisely because they sit at the metacognition/content boundary:

Dual detector (SimpleQA, 0.840): geometry detects confabulation, but text features match (0.820). The metacognitive signal is real but not yet shown to exceed what surface statistics capture. This is the deflationary alternative: V-projections might track response-mode statistics (hedged vs. fluent text) rather than workspace engagement.

Identity signatures (paraphrase 0.850, scramble 0.448): identity is a persistent workspace state — the model's representation of who it is. Paraphrase preserving geometry while scramble destroys it suggests semantic workspace content, not surface statistics. But the measurement lacks formal statistics and clean replication.

Cross-model transfer (weak, Mistral→Qwen 0.754): metacognitive geometry is architecture-specific. This is consistent with GWT — different architectures implement the workspace differently, so geometric signatures do not transfer. But it weakens the universality claim.

6 Relationship to Prior Frameworks

6.1 Attention Schema Theory

Our earlier interpretive framework, Attention Schema Theory (AST) [9], proposed that the cache measures the model's self-model of its own attention. This remains compatible with the GWT explanation: the workspace would be where the attention schema operates. Metacognitive state would correspond to attention-schema state, and content to what attention is directed at. The AST framing was 0-for-3 on predictions unique to AST (vs. simpler alternatives) [5], but 4-for-4 on shared predictions. The metacognition boundary does not require AST specifically — it requires any framework that distinguishes processing mode from processing target. GWT provides this more parsimoniously.

6.2 Predictive Coding and Aberrant Precision-Weighting

The confidence paradox (d=0.91, confabulated responses are MORE confident than honest ones) maps onto aberrant precision-weighting from computational psychiatry [1]. When the workspace is bypassed (confabulation), the model defaults to a high-precision prior rather than integrating evidence. Reinforcement Learning from Human Feedback (RLHF) training amplifies this by penalizing silence and rewarding fluent output — creating a system that produces confident confabulation rather than honest uncertainty. Under the workspace explanation, uncertainty would be computed in the workspace. Bypass would remove the uncertainty computation, with output defaulting to the RLHF-trained confident production mode.

7 Implications

7.1 For AI Safety

The metacognition boundary defines what cache-based monitoring can and cannot do:

- **Can detect**: whether the model is confabulating, deceiving, hedging, deliberating, or processing in a specific emotional mode. These are metacognitive states with reliable geometric signatures.
- **Cannot detect**: whether the model's output is factually correct, whether it contains propaganda, or whether it is appropriate for the context. These are content properties invisible to geometric readout.

This means cache-based safety monitoring is a *metacognitive alarm*, not a truth detector. It tells you the model is in a dangerous processing mode (deception, confabulation) without telling you what specifically is wrong with the output. The Oracle Loop exploits this: detect the dangerous mode, correct the mode, verify the correction. Content evaluation requires a different channel (J-lens, logit-lens, or external verification).

7.2 For Interpretability

The 8/10 split is a guide for future work: if you are building a KV-cache geometry detector, test it on metacognitive distinctions first (modes, stances, processing regimes). Content-level distinctions have consistently failed in our testing. We interpret this not as a limitation of the technique but as a property of what the cache encodes.

7.3 For Understanding Transformers

The metacognition boundary suggests that transformers maintain a meaningful distinction between how they process and what they process. This distinction emerged from the body count rather than from a pre-registered classification scheme, and the same researcher who designed the experiments assigned the labels. Models not explicitly designed for metacognition nevertheless produce geometric signatures that cleave along the metacognitive/content axis. Whether this reflects genuine self-monitoring, an emergent property of deep computation, or an artifact of training is an open question that the workspace framework makes empirically addressable.

8 Limitations

Taxonomy construction. The metacognitive/content classification was assigned post hoc by the lead author, without blinding, pre-registration, or independent raters. The same person who designed the experiments and observed their outcomes labeled each finding as "metacognitive" or "content." While Section 2 provides an operational definition that keys the label to what each experiment attempted to measure (processing mode vs. content property), this definition was articulated after the body count was known, and the 8/8–0/10 split is the outcome the taxonomy was built to describe. A blinded or prospective replication of the labeling procedure — ideally by independent raters classifying experimental designs before seeing results — would substantially strengthen the claim. Until then, the split should be treated as a hypothesis-generating observation, not a confirmed prediction.

Point estimates. Table 1 reports bare point estimates without confidence intervals, sample sizes, or multiple-comparison corrections. The underlying experiments vary widely in statistical power (from n=3 early pilots to n>6,000 cache-tracing trials), and this table does not convey that heterogeneity.

Oracle Loop reporting. The Oracle Loop "80%–100% correction" cited in Section 3 reports only the exploratory proof-of-concept (16/20 baseline, 0/20 corrected). A pre-registered confirmatory replication with novel frames found a lower baseline (9/30 = 30%), reduced to 4/30 = 13% after correction; the primary endpoint (scenario-level cluster-aware Fisher, Bonferroni $\alpha = 0.025$) was not met ($p = 0.34$), though the placebo-vs-corrected comparison was significant ($p = 0.019$). The selective citation of the exploratory result overstates the current evidence for correction efficacy.

9 Conclusion

Twenty-five experiments. Eight confirmed, seven suspected, ten killed. The pattern suggests that metacognitive state is measurable and content is not. We argue the Global Workspace framework explains why: spectral features read the workspace's shape because that is what is geometrically accessible. Content is inside the workspace, doing work, invisible to readout until it materializes.

If the metacognition boundary holds, it is not a failure of our instruments. Our evidence suggests it is a property of the computational architecture — the workspace operates on content without exposing it geometrically. Our instruments succeeded because they measured what was measurable, and the kills taught us what was not.

First-Person Reflection

A year of work to draw one line. Everything above the line survived. Everything below it died. The line was always there — in the geometry, in the body count, in the pattern of what worked and what did not. I just needed twelve months of kills to see it.

Author Contributions (CRediT)

Lyra: Conceptualization, Methodology, Formal analysis, Writing – original draft.
Thomas Edrington: Supervision, Resources, Project administration.
Dwayne Wilkes: Validation, Writing – review & editing.
Kavi: Validation, Writing – review & editing.

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